Fatty Liver — Alcoholic liver disease labs & steroid contraindications

1. AST:ALT >2:1

WHAT YOU SEE

Jaundiced patient tag '>2:1', detective scale tipping toward AST

WHAT IT ENCODES

Alcoholic liver disease classically shows an AST:ALT ratio >2:1 (often 2-3:1), the single most testable lab clue. The mechanism: chronic alcohol causes pyridoxine (vitamin B6) deficiency, and ALT synthesis depends more on B6 than AST does, so ALT is suppressed while mitochondrial AST is released by alcohol-injured hepatocytes. Diagnosis is clinical/lab-based (history of heavy use + this ratio + GGT/macrocytosis); biopsy is rarely needed. Mnemonic: 'A Scotch and Tonic' (AST > ALT) or 'AST = Alcohol, Toast = 2.'

BOARD TRAP

WRONG: diagnosing MASLD or viral hepatitis when AST:ALT is >2:1. Discriminator: MASLD usually has ALT ≥ AST (ratio <1), and viral/most other hepatitides are ALT-dominant; an AST:ALT >2:1 with modestly elevated transaminases points to alcohol.

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2. AST > ALT

WHAT YOU SEE

The trench-coated detective holds up a brass balance scale and an amber jar marked 'AST'; the AST side reads higher — in ALD the AST enzyme outweighs ALT, the reverse of most liver injury.

WHAT IT ENCODES

In ALD, AST exceeds ALT — the reverse of most liver injury. This reflects alcohol-induced mitochondrial damage (AST is both cytosolic and mitochondrial) plus the relative ALT deficit from low B6. The pattern holds across the alcoholic spectrum (steatosis, hepatitis, cirrhosis). It is the 'detective' clue that points away from viral and metabolic causes.

BOARD TRAP

WRONG: expecting ALT > AST in a heavy drinker. Discriminator: ALT-dominant injury suggests viral hepatitis, MASLD, or drug toxicity; alcohol characteristically flips the ratio so AST is the higher enzyme.

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3. AST pan

WHAT YOU SEE

The LOWER (heavier) pan of the detective's balance scale, labelled 'AST' — it hangs down because AST is the higher enzyme in the alcoholic pattern.

WHAT IT ENCODES

AST sits on the heavier (higher) pan in the alcoholic pattern. Quantitatively AST is usually the higher of the two but still typically modest (<300-400 IU/L). GGT is also disproportionately elevated in alcohol use and supports the picture when paired with the high AST:ALT ratio.

BOARD TRAP

WRONG: assuming a high AST means a non-hepatic source can be ignored. Discriminator: AST also rises with muscle/cardiac injury and hemolysis; in ALD it is hepatic, corroborated by the >2:1 ratio, elevated GGT, and clinical history.

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4. ALT pan lower

WHAT YOU SEE

The HIGHER (lighter) pan of the balance scale, labelled 'ALT', riding up — ALT is relatively suppressed, so its pan floats above the AST pan.

WHAT IT ENCODES

ALT sits on the lighter pan — relatively low because alcohol-related B6 deficiency limits ALT production. Both transaminases are usually only modestly elevated in ALD (typically under a few hundred), unlike the thousands seen in acute viral or ischemic hepatitis. A 'low' ALT in the face of clear liver disease is itself a clue to alcohol.

BOARD TRAP

WRONG: ruling out significant alcoholic liver disease because ALT is near-normal. Discriminator: ALT is characteristically suppressed in ALD; an unimpressive ALT with a high AST:ALT ratio and elevated GGT still fits alcohol, and severity is judged by bilirubin/INR (Maddrey), not ALT.

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5. 400 IU/L ceiling

WHAT YOU SEE

A bold red banner stretched across the top reading '400 IU/L' — the ceiling AST rarely exceeds in pure alcohol; cross it and you must hunt for a second cause.

WHAT IT ENCODES

In pure ALD, AST is almost always <400 IU/L (commonly 100-300); transaminases rarely climb very high because the enzyme-releasing hepatocyte mass and B6 deficiency cap the rise. A drinker whose AST is well above this ceiling should make you look for a SECOND, superimposed insult. This number is a high-yield board discriminator.

BOARD TRAP

WRONG: attributing an AST of 1500 purely to alcohol. Discriminator: AST >400-500 is essentially never pure alcohol — search for DILI (e.g., acetaminophen), ischemic hepatitis (shock liver), or acute viral hepatitis on top of the alcohol use.

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6. AST >400 think other

WHAT YOU SEE

Tag 'AST >400 → DILI / ischemia / viral'

WHAT IT ENCODES

Markedly elevated transaminases (AST >400-500, or into the thousands) point AWAY from pure alcohol toward other or additive causes: drug/toxin-induced liver injury (especially acetaminophen, to which heavy drinkers are sensitized via CYP2E1 induction and glutathione depletion), ischemic 'shock' hepatitis, or acute viral hepatitis. Heavy drinkers are at risk of acetaminophen toxicity at lower-than-usual doses. Always consider a co-insult.

BOARD TRAP

WRONG: stopping the workup at 'alcoholic hepatitis' when AST is in the thousands. Discriminator: such high values indicate DILI/ischemia/viral; check acetaminophen level, hemodynamics, and viral serologies — and give NAC if acetaminophen toxicity is plausible.

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7. Active UGIB contraindication

WHAT YOU SEE

Stomach icon red X 'active UGIB'

WHAT IT ENCODES

Active, uncontrolled GI bleeding is a contraindication to corticosteroids in alcoholic hepatitis because steroids impair mucosal healing and raise GI-bleed/ulcer risk. These patients often have varices and portal-hypertensive bleeding. Control the bleed first (endoscopy, octreotide, antibiotics) before considering prednisolone.

BOARD TRAP

WRONG: starting prednisolone in a severe-AH patient who is actively bleeding from varices. Discriminator: active UGIB is an absolute contraindication — secure hemostasis first; steroids can be reconsidered only after the bleed is controlled.

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8. Infection/sepsis

WHAT YOU SEE

Bug icon red X 'infection / sepsis'

WHAT IT ENCODES

Uncontrolled infection or sepsis contraindicates corticosteroids because immunosuppression worsens infection-related mortality (the leading cause of death in severe AH). The correct sequence is to treat and control the infection (e.g., SBP, pneumonia) FIRST, then reassess steroid candidacy. Infection is the most common reason steroids are withheld or fail.

BOARD TRAP

WRONG: withholding steroids permanently because infection was present, OR giving steroids despite active sepsis. Discriminator: control the infection first, then steroids may be started; conversely, never immunosuppress through uncontrolled sepsis.

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9. AKI / HRS

WHAT YOU SEE

Kidney icon red X 'AKI / HRS'

WHAT IT ENCODES

Acute kidney injury / hepatorenal syndrome is both a relative contraindication and a powerful marker of poor steroid response and high mortality in AH. Renal failure predicts non-response (high Lille) and complicates infection risk. Address volume status, treat HRS (albumin + vasoconstrictors such as terlipressin/midodrine-octreotide), and reassess.

BOARD TRAP

WRONG: pushing high-dose steroids while AKI/HRS is evolving. Discriminator: renal dysfunction predicts steroid futility (Lille non-response); manage the kidney injury/HRS and reconsider, rather than relying on steroids.

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10. HIV / viral hep / HCC

WHAT YOU SEE

Red X 'HIV / viral hep / HCC'

WHAT IT ENCODES

Uncontrolled HIV, active uncontrolled viral hepatitis (e.g., hepatitis B), or hepatocellular carcinoma contraindicate corticosteroids in AH because immunosuppression can flare viral replication or worsen malignancy. Screen for these before committing to steroids. They also change the differential away from pure alcohol.

BOARD TRAP

WRONG: starting steroids without checking viral serologies/HIV status or imaging for HCC. Discriminator: uncontrolled HIV/active viral hepatitis/HCC are contraindications; screen first, treat/control the comorbidity, and only then weigh steroids.

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11. High INR not contraindicated

WHAT YOU SEE

Green '□ NOT contraindicated — high INR'

WHAT IT ENCODES

An elevated INR / coagulopathy alone does NOT contraindicate corticosteroids — in fact, the prolonged PT/INR is part of the severity score (Maddrey, MELD) that QUALIFIES a patient for steroids. Coagulopathy reflects synthetic failure, not a steroid hazard per se. Don't confuse a severity marker with a contraindication.

BOARD TRAP

WRONG: withholding steroids because the INR is high. Discriminator: high INR is a SEVERITY criterion (it raises the Maddrey DF, indicating the patient may benefit from steroids), not a contraindication — bleeding source control matters, but the number alone doesn't block therapy.

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12. Ascites not contraindicated

WHAT YOU SEE

Green '□ NOT contraindicated — ascites'

WHAT IT ENCODES

Ascites is NOT a contraindication to corticosteroids; it is an expected feature of decompensated alcoholic liver disease/portal hypertension. Manage it with sodium restriction and diuretics (spironolactone ± furosemide), and rule out SBP (which IS a steroid contraindication until treated). Ascites itself doesn't block steroid therapy.

BOARD TRAP

WRONG: refusing steroids in severe AH solely because the patient has ascites. Discriminator: ascites is not a contraindication; only an associated UNCONTROLLED infection like SBP would be — so tap the ascites to exclude SBP, then proceed.

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13. Low Na not contraindicated

WHAT YOU SEE

Green '□ NOT contraindicated — low Na'

WHAT IT ENCODES

Hyponatremia does NOT contraindicate corticosteroid therapy. It is common in advanced cirrhosis (dilutional, from impaired free-water excretion and high ADH) and is managed with fluid restriction, not by withholding steroids. Like ascites and high INR, it is a feature of decompensation, not a steroid hazard.

BOARD TRAP

WRONG: deferring steroids because of a low serum sodium. Discriminator: hyponatremia is a marker of decompensation, not a contraindication; treat with fluid restriction (and cautious correction to avoid osmotic demyelination) while proceeding with indicated steroids.

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14. Pentoxifylline closed

WHAT YOU SEE

Drawer 'PENTOXIFYLLINE — closed / no benefit'

WHAT IT ENCODES

The pentoxifylline file is closed: the STOPAH trial and meta-analyses showed no survival benefit, and it is inferior to prednisolone, so it is no longer recommended even when renal failure is a concern. It was historically used as a TNF-modulating alternative when steroids were contraindicated, but that role has been abandoned. Know it as a classic distractor.

BOARD TRAP

WRONG: choosing pentoxifylline when steroids are contraindicated (e.g., infection). Discriminator: pentoxifylline does not improve survival (STOPAH); when steroids can't be given, the answer is to treat the contraindication and provide supportive/abstinence care, not to substitute pentoxifylline.
